

**US Geological Survey
Geology Minerals Energy and Geophysics
Statement of Work**

1.0 INTRODUCTION

The USGS needs to acquire structural geology software capable of data integration, forward and reverse cross-section modelling, and kinematic restoration and validation.

2.0 BACKGROUND

The USGS conducts research on regional geology, geophysical, and metallogenic framework studies that elucidate the distribution of critical mineral resources in the United States. Structural and flexural-kinematic modelling can provide constraints on the order and magnitude of deformation, the tectonic evolution, and through integration with thermal models, time and temperature history. The time-temperature the time-depth history from modelling provides critical information for constraining the tectonic evolution and conceptual models for ore-forming processes. To complete this type of modelling, we need structural geology software capable of data integration, forward and reverse cross-section modelling, and kinematic restoration and validation. This software will help us constrain the geologic evolution and time-temperature/time-depth history of the northeastern Washington region and will ultimately help us determine the location and formation history of potential critical minerals.

3.0 SCOPE

The USGS requires access to software that can import, scale, and georeference 2-D cross-sections, forward and reverse model fault motion, uplift, loading, erosion, unloading, sediment accumulation, and sediment loading. The software must be able to assign different densities and physical properties of the mantle and crustal rocks, and must be able to create, record, and export large point clouds for every step of fault motion, uplift, loading, erosion, unloading, sediment accumulation, and sediment loading through the model. These point clouds are the product we integrate into thermal models and are central to the project, and the overall critical mineral potential modelling in northeastern Washington State

4.0 APPLICABLE DOCUMENTS N/A

5.0 TECHNICAL REQUIREMENTS

The software needs to be capable of importing and integrating data types such as digital field data, digital elevation models, seismic data, well and borehole data, geological maps, point clouds and point data, volume data, annotated field images, scanned cross-sections, gravity/magnetic and remotely sensed data, voxel, ASCII, GIS shape and DXF files. The software must have tools to restore and forward model cross-sections and specifically include a displacement analysis and fault validity testing module. The software must also be capable of

modelling block restoration, flexural slip, simple shear, trishear, fault parallel flow, fault bend folding, and elliptical fault flow. The contractor shall have distribution rights for the latest version of the software and be able to provide the software and associated license to the software.

6.0 DELIVERABLES

The contractor shall provide the structural geology modelling software and the USB key license to the USGS. The contractor shall also provide annual maintenance of the software and technical support.

7.0 PERIOD OF PERFORMANCE: One year from the date of award and two Option years until December 31, 2029.